Mental Health Analysis

A Project Report

submitted in partial fulfillment of the requirements

of

Foundation Course

by

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ACKNOWLEDGEMENT

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A special thanks to the institutions, organizations, and individuals who provided the dataset and resources essential for this research. Without their support, the analysis and findings would not have been possible.

Lastly, We are grateful to our family and friends for their patience, motivation, and encouragement during the course of this project. Their unwavering belief in our abilities kept us motivated even during hard challenges.

This project has been an enriching learning experience, and we deeply appreciate the knowledge and skills we have gained throughout this journey.

ABSTRACT

Mental health analysis is a crucial step in understanding the factors that contribute to an individual's well-being. This study delves into various lifestyle, demographic, and health-related factors to identify patterns that may influence mental health conditions, particularly depression. By analyzing key indicators such as sleep patterns, dietary habits, physical activity, and medical history, we aim to uncover correlations that could help in early detection and prevention strategies. The findings from this analysis can support healthcare professionals, policymakers, and researchers in developing more effective interventions and awareness programs to promote mental well-being in society.

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INTRODUCTION

CHAPTER 1

INTRODUCTION

Problem Statement: The goal is to analyze a depression dataset to identify trends and predictors of depressive symptoms, assess the effectiveness of current interventions, and uncover disparities in mental health care access. This analysis aims to provide actionable insights for improving prevention strategies, enhancing treatment accessibility, and reducing the overall burden of depression on public health.

Problem Definition: Depression is a complex condition influenced by multiple factors, including lifestyle, personal experiences, and societal conditions. Traditional approaches to diagnosing depression rely on self-reported symptoms and clinical evaluations, which may not always be timely or accurate. With the rise of data analytics, we now have the opportunity to analyze real-world datasets and uncover patterns that could provide early warning signs. This project focuses on analyzing a dataset related to depression to extract meaningful insights, helping researchers and mental health professionals make informed decisions

1.3 Expected Outcomes: By conducting an in-depth analysis of depression-related data, we aim to achieve the following:

- **Understanding Key Trends**: Identifying common factors associated with depression, such as age,employment, sleep patterns and lifestyle choices.
- **Data-Driven Insights**: Finding correlations between various aspects of mental health and external influences.
- **Support for Mental Health Awareness**: Providing insights that can be used to educate people and encourage early intervention.

1.4 Organization of the Report

The remaining report is organized as follows:

Chapter 2 : A review of existing research and literature on depression analysis using data science.

Chapter 3 : Details about the dataset used and implementation preprocessing techniques, feature selection , visualization.

Chapter 4 : Discussion of findings, results, and their significance in understanding depression.

Chapter 5 : Conclusion, benefits , future work and references.

CHAPTER 2

LITERATURE SURVEY

CHAPTER 2

LITERATURE SURVEY

2.1. Paper-1: "Analyzing mental health services data"

2.1.1 Brief Introduction of Paper :

- In mental health services research, analyzing service utilization data often poses serious problems, given the presence of substantially skewed data distributions. This article presents a non-technical introduction to statistical methods specifically designed to handle the complexly distributed datasets that represent mental health service use, including Poisson, negative binomial, zero-inflated, and zero-truncated regression models. A flowchart is provided to assist the investigator in selecting the most appropriate method. Finally, a dataset of mental health service use reported by medical patients is described, and a comparison of results across several different statistical methods is presented. Implications of matching data analytic techniques appropriately with the often complexly distributed datasets of mental health services utilization variables are discussed.

2.1.2 Techniques used in Paper :

- Statistical correlation analysis and trend evaluation.
- Count Regression Model Applied

2.2. Paper-2: "Mental Health Prediction using Sentiment Analysis”

2.2.1 Brief Introduction of Paper :

- This paper proposes a system that leverages sentiment analysis of text data to predict mental health issues. By analyzing the sentiments and expressions conveyed in textual inputs, the system aims to assess the likelihood of individuals experiencing conditions like anxiety, depression, or stress.

2.2.2 Techniques used in Paper :

- The approach utilizes natural language processing techniques to process text data, including tokenization, text cleaning, part-of-speech tagging, and lemmatization.
- Machine learning algorithms are then applied to classify the sentiment and predict potential mental health issues based on the analyzed text.

2.3. Paper-3: "Prediction of Mental Health Based on Data Science”

2.3.1 Brief Introduction of Paper :

- This study focuses on developing predictive models to identify individuals at risk of mental health issues. By analyzing a large dataset encompassing demographic information, lifestyle factors, and clinical symptoms, the research aims to enhance early detection and intervention strategies for mental health disorders.

2.3.2 Techniques used in Paper :

- Data pre-processing, feature selection, and machine learning models, to analyze variables related to mental health
- Machine Learning models using various algorithms, including decision trees, random forests, and support vector machines, to predict mental health issues based on the processed data.

CHAPTER 3

PROPOSED METHODOLOGY

CHAPTER 3

PROPOSED METHODOLOGY

3.1 System Design

The system aims to analyze depression-related data using statistical methods and exploratory data analysis (EDA). The workflow includes :

3.1.1 Data Collection & Preprocessing

- The dataset consists of responses from individuals regarding mental health, work-life balance , employment , sleep patterns , previous medical records , alcohol consumption , dietary habits , etc.
- Data Cleaning is performed to remove missing or inconsistent values.
- Feature Selection ensures that only the most relevant attributes are analyzed.

3.1.2 Feature Engineering & Statistical Analysis

- Key features such as sleep patterns, dietary habits , employment , and physical activity are analyzed to detect patterns linked to depressive symptoms.
- Correlation analysis helps identify dependencies between factors (e.g., does lower sleep correlate with higher depression?).

3.2.1 Data Analysis & Visualization

The system employs statistical tools for analysis and visualization, including:

- Descriptive Statistics to summarize depression trends across demographics.
- Data Visualization using histograms, and bar charts for deeper insights.

3.2 Data Flow Diagram

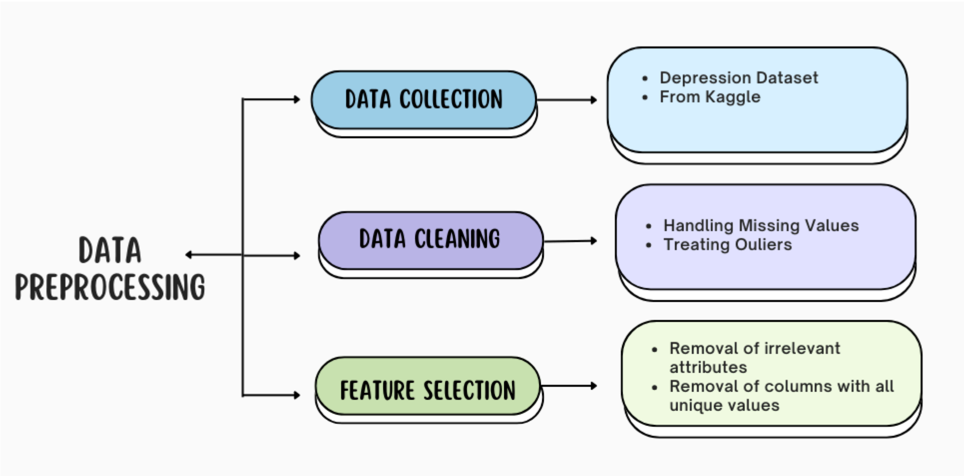
A Data Flow Diagram (DFD) is a graphical representation of the "flow" of data through an information system, modeling its process aspects. A DFD is often used as a preliminary step to create an overview of the system, which can later be elaborated. DFDs can also be used for the visualization of data processing (structured design).

3.3.1 DFD Level 0 – Overview

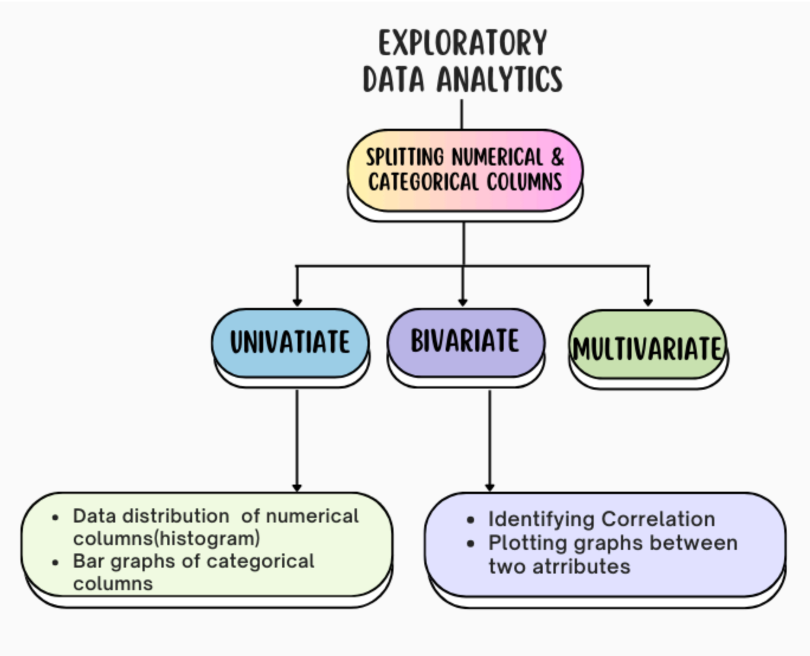
This diagram gives a broad view of the **data processing pipeline**, from input collection to analysis.

3.3.2 DFD Level 1 – Data Processing Modules
The system is divided into three primary modules:

- **DFD Level 1 - Data Collection & Cleaning:** Handles missing data and inconsistencies.
- **DFD Level 1 - Feature Engineering :** Remove unique and relevant attributes.



3.3.3 DFD Level 2 – Exploratory Data Analysis
- **DFD Level 2 - Statistical Analysis Module:** Computes mean, median, standard deviation, distribution and correlations.
- **DFD Level 2 - Visualization Module:** Generates histograms and bar graphs.



3.4 Advantages

Our proposed system offers several advantages that enhance its usability and effectiveness:

- **Effective Data Visualization** – Offers clear graphical representations for better understanding.
- **Enhance Research and Understanding** - Can use these insights in mental health related research.
- **ML Implementation** - Analysis can be used for ML model fitting.
- **Predictive analysis** - can perform predictive analysis on data in order to provide early detection and intervention.

3.5 Requirement Specification

To ensure smooth implementation and execution, the system requires specific hardware and software configurations.

3.5.1 Hardware Requirements

- Processor: Intel Core i5 or higher
- RAM: Minimum 8GB (16GB recommended for better performance)
- Storage: At least 256GB SSD (HDD compatible but slower)
- GPU (if applicable): NVIDIA GTX 1050 or equivalent for advanced processing

3.5.2 Software Requirements

- Operating System: Windows 10/11, Linux (Ubuntu), or macOS
- Programming Language: Python 3.x
- Libraries: Pandas, NumPy, Matplotlib, Seaborn (for data analysis and visualization)
- Frameworks: Jupyter Notebook for data processing
- Database: Depression.csv dataset

CHAPTER 4

Implementation and Result

CHAPTER 4

IMPLEMENTATION and RESULT

4.1 Introduction

This chapter presents the implementation details and results obtained from the analysis of the dataset. The main objective is to extract meaningful insights related to depression analysis by applying various data processing, visualization, and EDA techniques. We will go through the data preprocessing steps and exploratory data analysis .

4.2 Dataset Overview

The dataset used in this project consists of **100,000 observations** with **16 attributes (columns)**. The attributes include a mix of **numerical and categorical** variables, which capture different aspects related to depression analysis. Before proceeding with the implementation, an initial exploration of the dataset was conducted to understand its structure, detect missing values, treating inconsistent values and prepare it for further analysis.

About the Dataset:

- The dataset contains **3 numerical** and **13 categorical** columns.
- No missing values were found in the dataset.
- One column with **all unique values** i.e **Name**.
- Have **outliers in income** attributes.
- The dataset size is **12.2 MB**, meaning it does not require extra computational resources.

4.3 Data Preprocessing

Data preprocessing is a crucial step in data analysis that involves transforming raw data into a clean, consistent, and usable format. It is essential for ensuring the accuracy, completeness, and reliability of analytical results.

1. **Data Cleaning:** Since no missing values were found, no imputation was required.
2. **Treating Outliers :** 'Income' attributes have outliers , which was treated by replacing them by mean values.
3. **Categorical Data Encoding :** Categorical features were transformed into numerical representations using encoding techniques.
4. **Feature Selection:** Features were examined to determine their relevance , where the Name attribute was removed as it has all the unique values.

- ## Column Information Table :

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100000 entries, 0 to 99999
Data columns (total 16 columns):
 #   Column                        Non-Null Count   Dtype  
---  ------                        --------------   -----  
 0   Name                          100000 non-null  object 
 1   Age                           100000 non-null  int64  
 2   Marital Status                100000 non-null  object 
 3   Education Level               100000 non-null  object 
 4   Number of Children            100000 non-null  int64  
 5   Smoking Status                100000 non-null  object 
 6   Physical Activity Level       100000 non-null  object 
 7   Employment Status             100000 non-null  object 
 8   Income                        100000 non-null  float64
 9   Alcohol Consumption           100000 non-null  object 
 10  Dietary Habits                100000 non-null  object 
 11  Sleep Patterns                100000 non-null  object 
 12  History of Mental Illness     100000 non-null  object 
 13  History of Substance Abuse    100000 non-null  object 
 14  Family History of Depression  100000 non-null  object 
 15  Chronic Medical Conditions    100000 non-null  object 
dtypes: float64(1), int64(2), object(13)
memory usage: 12.2+ MB
```

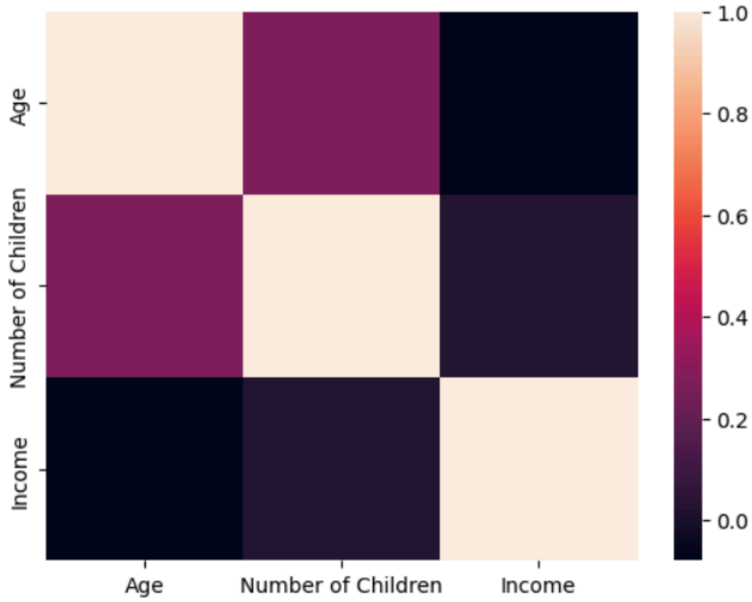
- ## Numerical Column Description Table :

	count	mean	std	min	25%	50%	75%	max
Age	100000.0	49.093030	18.190601	18.00	33.0000	49.000	65.00	80.00
Number of Children	100000.0	1.299620	1.236205	0.00	0.0000	1.000	2.00	4.00
Income	100000.0	50595.977933	40510.360823	0.41	21028.8225	37545.425	76283.42	209995.22

- **Correlation Table :**

	Age	Number of Children	Income
Age	1.000000	0.267108	-0.079224
Number of Children	0.267108	1.000000	0.025115
Income	-0.079224	0.025115	1.000000

<Axes: >



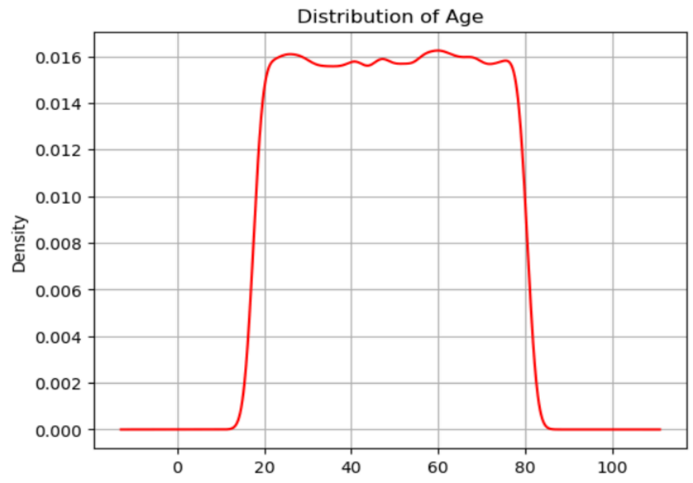
4.4 Exploratory Data Analysis (EDA)

EDA was conducted to uncover patterns and trends in the dataset. Several visualizations were used:

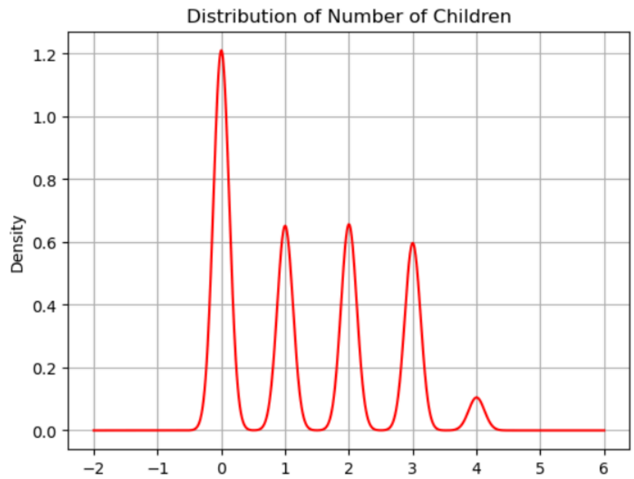
- **Univariate Analysis:** Individual features were analyzed to understand their distribution.
- **Bivariate Analysis:** Relationships between variables were explored using bar graphs.
- **Heatmaps:** A heatmap was used to visualize the correlation between numerical features.

4.5 Results and Findings :

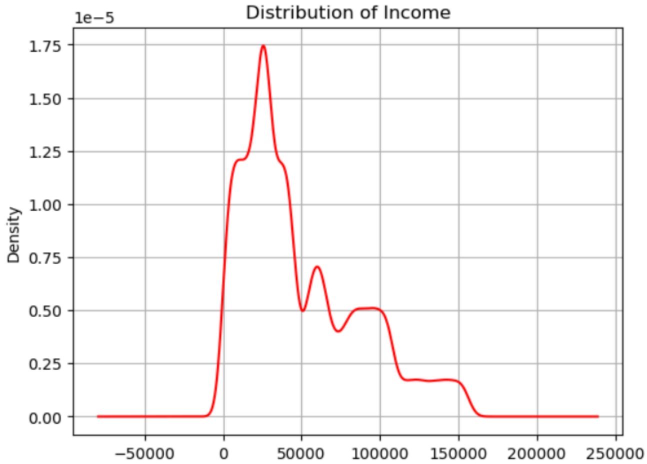
- Distribution of numerical data
  1) Age
     - Minimum age of the person is 18.
     - Maximum age of the person is 80.
     - Average age of the person is 49.09.
     - The data follows the multimodal distribution (range is from 20 to 80)



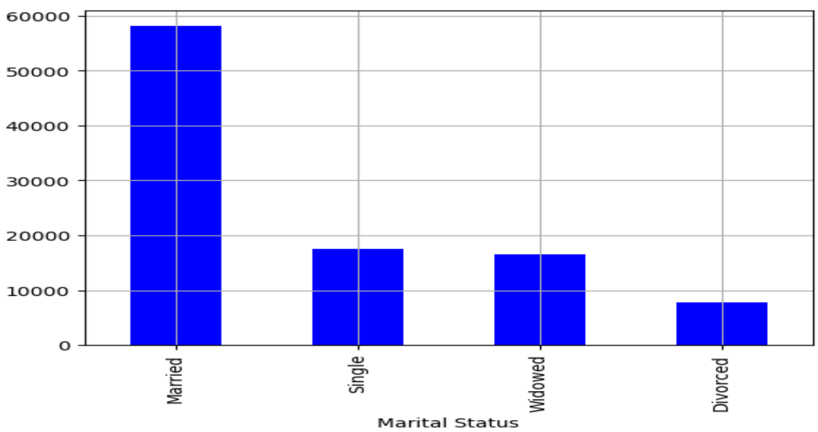
  2) Number of Children
     - Minimum number of children a person have is 0
     - Maximum number of children a person have is 4
     - Average of a person having number of children is around 2
     - The data follows multimodal distribution
     - The data have discrete values of ranges between 0 to 4.

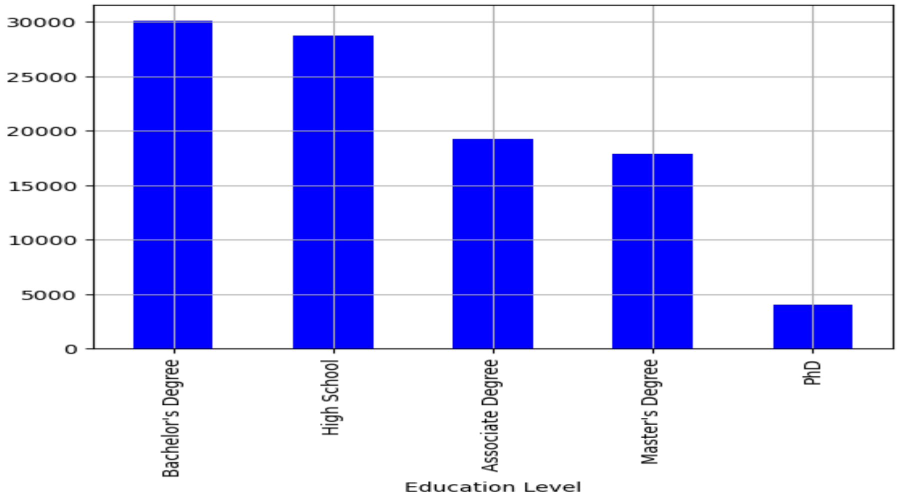


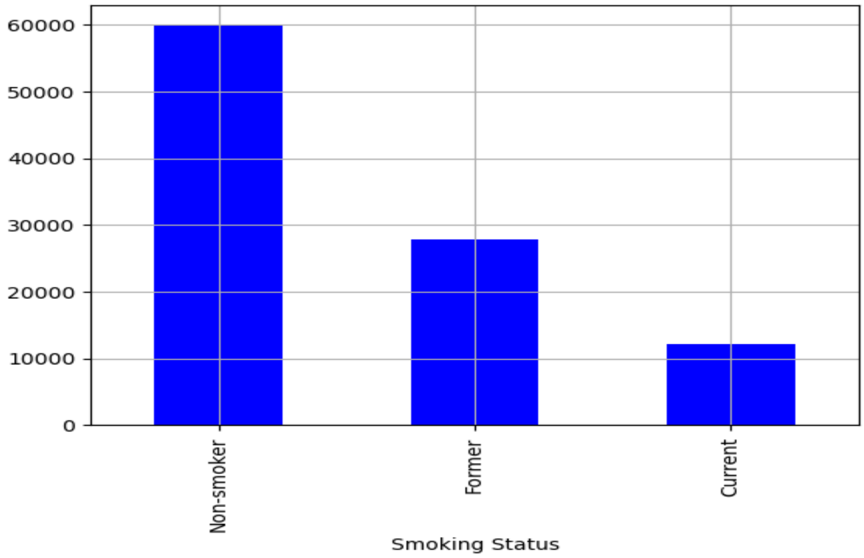
3) Income
    - Minimum income of people is 0.41.
    - Maximum income of people is 209,995.
    - Average income of people is 50,595.
    - The Data is right skewed and multimodal.

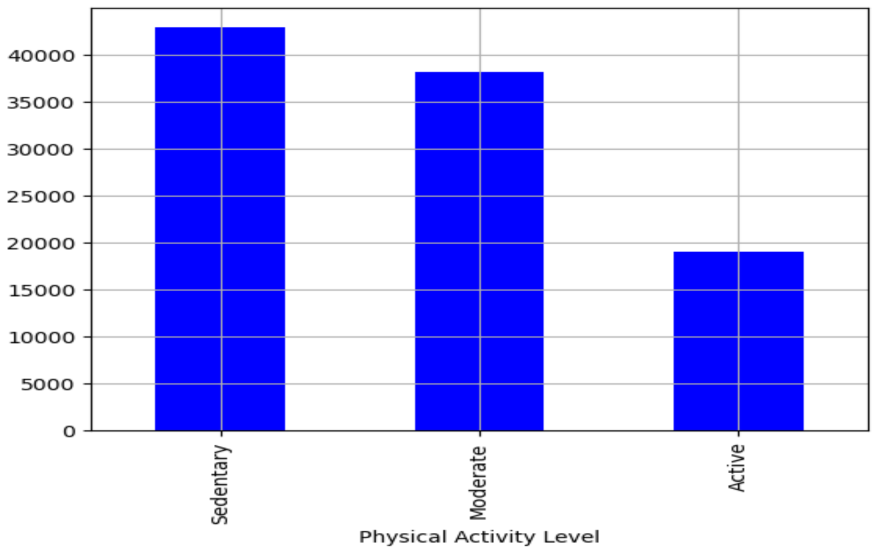


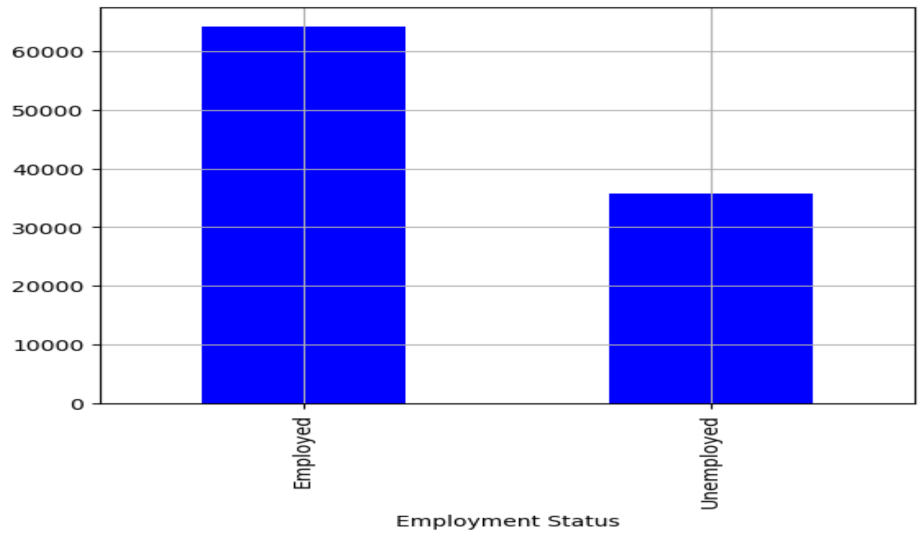
- Visualization of categorical column

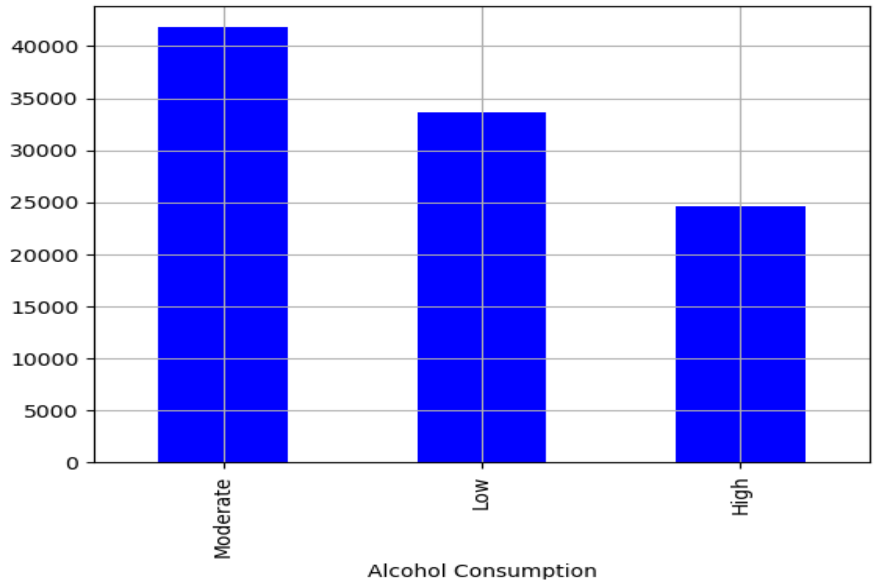


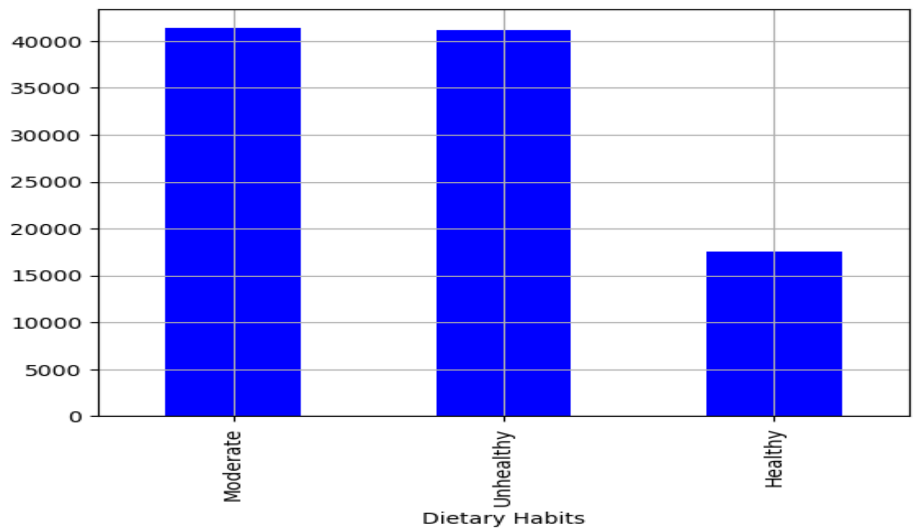

30000
25000
20000
15000
10000
5000
0
Bachelor's Degree
High School
Associate Degree
Master's Degree
PhD
Education Level

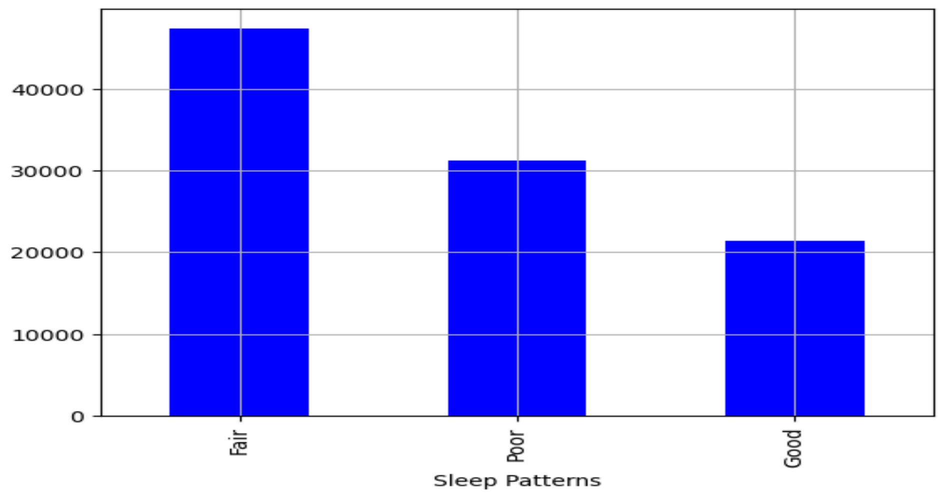

60000
50000
40000
30000
20000
10000
0
Non-smoker
Former
Current
Smoking Status


40000
35000
30000
25000
20000
15000
10000
5000
0
Sedentary
Moderate
Active
Physical Activity Level


60000
50000
40000
30000
20000
10000
0
Employed
Unemployed
Employment Status


40000
35000
30000
25000
20000
15000
10000
5000
0
Moderate
Low
High
Alcohol Consumption


40000
35000
30000
25000
20000
15000
10000
5000
0
Moderate
Unhealthy
Healthy
Dietary Habits



INSIGHTS :

1. Possible age range to acquire depression is 18 to 80.
2. Married people have the highest probability of acquiring depression and divorced people have lower possibilities.
3. People with high school education and Bachelor degrees are the highest number of people with depression.
4. Majority of people are non-smokers.
5. People with sedentary physical activity are more prone to be depressed.
6. Employees are more depressed than unemployed people.
7. People with high alcohol consumption are the least depressed and moderate consumers are the most depressed
8. People with healthy diet have the least possibility to get depressed while people with unhealthy and moderate diet have highest possibility
9. People with Good sleep pattern have the least possibility to get depressed
10. People with no history of mental illness are the most depressed
11. People with no history of substance abuse are the most depressed
12. Number of people with no Family history of depression and no Chronic Medical Condition are high.

CHAPTER 5

CONCLUSION

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CONCLUSION

The results of this project assist medical professionals, researchers, and policymakers in making decisions about mental health awareness and treatment strategies. Moreover, it also helps professionals to bring AI/ML based solutions which leads to early intervention and detection strategies in the mental health domain.

5.1 Advantages

- **Effective Data Visualization** – Offers clear graphical representations for better understanding.

- **Enhance Research and Understanding** - Can use these insights in mental health related research.

- **AI/ML Implementation** - Analysis can be used for ML model fitting or AI implementation.

- **Predictive analysis** - can perform predictive analysis on data in order to provide early detection and intervention.

5.2 Scope for Future Work

Although the present project sets a solid ground for depression analysis, there remains vast scope for additional enhancements and extensions, including:

- **Integration with Real-World Data Sources:** Subsequent research can integrate real-time data from mental health mobile applications, surveys, or electronic medical records.
- **AI/ML Implementation** - Analysis can be used for ML model fitting or AI implementation.
- **Incorporating Deep Learning:** State-of-the-art deep learning architectures like transformers and LSTMs may be investigated to make more precise predictions.
- **Personalized Mental Health Recommendations:** AI-based personalized treatment could be established to offer personalized mental health interventions.

- **Cross-Cultural Analysis:** Increasing the size of the dataset to encompass multicultural populations may add to the results' generalizability.

GITHUB LINK :

https://github.com/KrishiPatel19/Data-Analysis-on-Depression-Dataset.git

VIDEO LINK :

https://drive.google.com/file/d/1X4G7LZD3flT4UzILQbMeEKJHN1JErz9Y/view?usp=drive_link

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2. Mental Health Prediction using Sentimental Analysis,Authors: Rishabh Verma, Nipun , Nitin Rana, Dr. Rakesh Kumar Arora, DOI Link: https://doi.org/10.22214/ijraset.2023.57534
3. Statistical procedures for analyzing mental health services data**,** Authors - Jon D. Elhai a, Patrick S. Calhoun b, Julian D. Ford c, DOI Link : https://doi.org/10.1016/j.psychres.2007.07.003
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